

Student Name:
University Roll No.:

Printed Pages: 02

School of Engineering
Second Theory Sessional Examination

B. Tech: CSE/CSE-A/CSE-IOTBC, CSE-CCML
Even Semester (AS: 2024-25)

Course Title: Design & Analysis of Algorithm
Course Code: BCSS3602 | CSE-AII
Year: 3I | Semester: VII
Max Marks: 30
Time: 1hrs

BAI3602 | CSE, CSE-CCML, CSE-IOTBC

Instructions: 1-Mention any assumptions made.
2-Notations have usual meaning.

SECTION 'A'
Q.N.1. Attempt all parts of the following:

Course Objective	Marks
a) Compare Greedy Programming, Dynamic programming & Divide & conquer.	CO3 1
b) Define Single source shortest path,	CO3 1
c) Give one difference between BFS and DFS	CO4 1
d) Define P, NP and NP-complete in decision problems	CO4 1
e) What do you mean by Approximation algo.	CO4 1

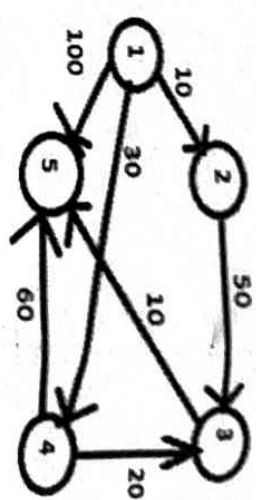
SECTION 'B'
Q.N.2. Attempt any two parts of the following:

Course Objective	Marks
a) Describe the Dynamic 0/1 Knapsack Problem. Find an optimal solution for the dynamic programming 0/1 knapsack instance for $n=3$, $m=6$, profits are $(p_1, p_2, p_3) = (1, 2, 5)$, weights are $(w_1, w_2, w_3) = (2, 3, 4)$.	CO3 7.5
b) Define backtracking? Solve N-Queens problem for $n=4$ showing backtracking steps.	CO3 7.5

c) Explain Rabin Karp string matching Algorithm. Simulate Rabin-Karp string matching for pattern "abc" in text "abcabcabc" with $q=101$.	CO4	7.5
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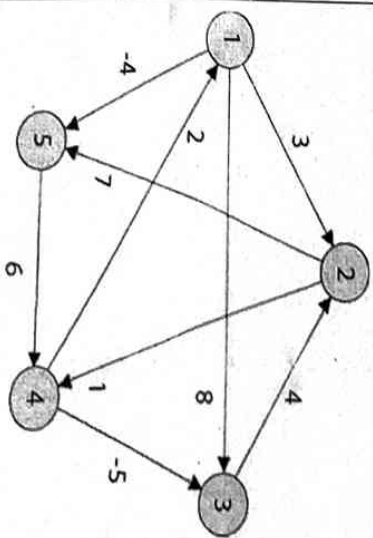
SECTION 'C'
Q.N.3. Attempt any one parts of the following:

a) Find the shortest path in the below graph from the source vertex 1 to all other vertices by using Dijkstra's algorithm.	CO4	10
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b)	CO5	10
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Explain Floyd warshell with an algorithm Find out all pair shortest path using Floyd warshell for given graph:



- c) A thief enters a house for robbing it. He can carry a maximal weight of 60 kg into his bag. There are 5 items in the house with the following weights and values. What items should thief take to get maximum benefit if he can even take the fraction of any item with him?

CO5

10

Item	Weight	Value
1	5	30
2	10	40
3	15	45
4	22	77
5	25	90

Table 1: Mapping between COs and questions

COs	Questions Numbers	Total Marks
CO3	1a, 1b, 1c, 2a, 2b	18
CO4	1d, 1e, 2c, 3a	19.5
CO5	3b, 3c	20

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06.05.2024